Enrollment No : 240840116062

Practical 6 : Create basic data visualizations (Line, Bar, Histogram, Scatter, Pie).

Step 1 :-Import the Libaries

```
# import  libraries
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns

import warnings
warnings.filterwarnings('ignore')
```

Step 2 :- Create a Dataset

```
# creating a dataset
data = {
    "Month" : ['Jan','Feb','Mar','Apr','May','Jun','Jul','Aug','Sep','Oct','Nov','Dec'],
    "Product" : ['Laptop','Mobile','Tablet','Laptop','Mobile','Tablet','Laptop','Mobile','Tablet','Laptop','Mobile','Tablet
    "Region" : ["North","South','East','West","North","South','East','West","North","South','East','West'],
    "Sales" : [120000, 135000, 150000, 145000, 160000, 175000, 180000, 170000, 190000, 210000, 230000, 250000],
    "Units_Sold" : [12, 30, 25, 14, 35, 30, 18, 38, 32, 21, 45, 40]
}
```

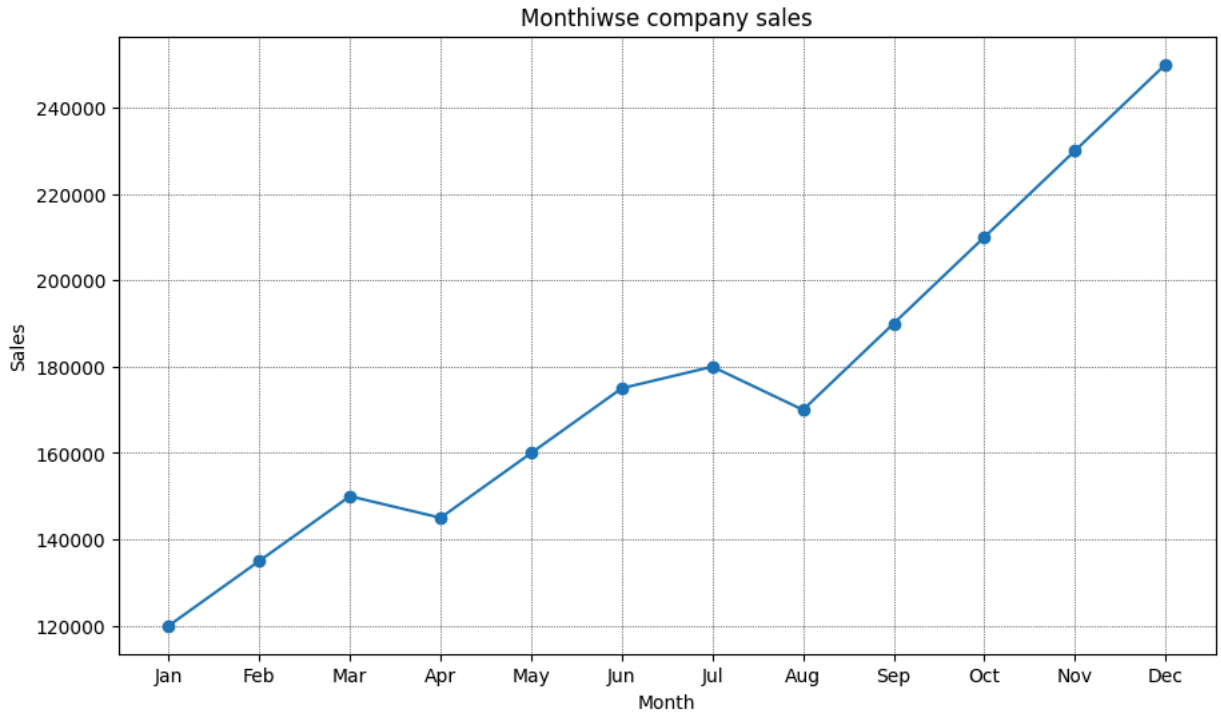
```
# create  dataframe from the sales.
df = pd.DataFrame(data)
df
```

	Month	Product	Region	Sales	Units_Sold
0	Jan	Laptop	North	120000	12
1	Feb	Mobile	South	135000	30
2	Mar	Tablet	East	150000	25
3	Apr	Laptop	West	145000	14
4	May	Mobile	North	160000	35
5	Jun	Tablet	South	175000	30
6	Jul	Laptop	East	180000	18
7	Aug	Mobile	West	170000	38
8	Sep	Tablet	North	190000	32
9	Oct	Laptop	South	210000	21
10	Nov	Mobile	East	230000	45
11	Dec	Tablet	West	250000	40

Next steps: Generate code with df | New interactive sheet

Step 3 : Create a line chart

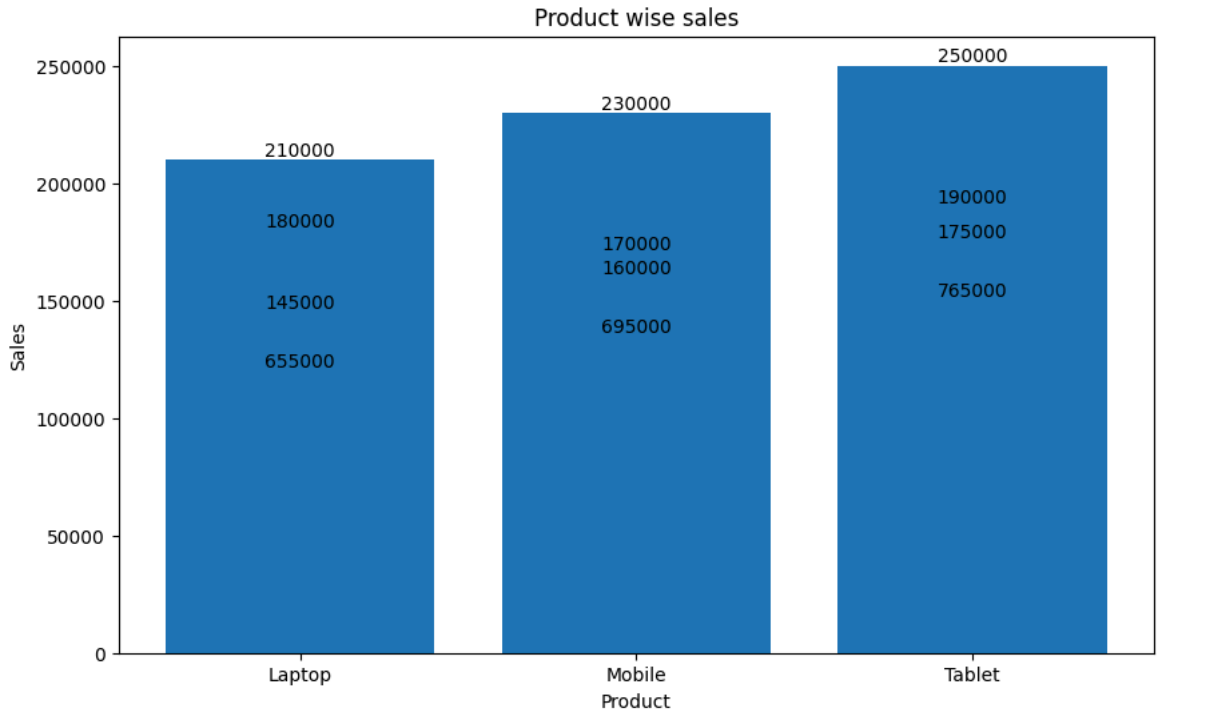
```
# create  line chart using company sales
plt.figure(figsize=(10.6, 6))
plt.plot(df['Month'], df['Sales'], marker='o')
plt.grid(True, which='both', linestyle=':', linewidth=0.5, color='black')
plt.title('Monthiwse company sales')
plt.xlabel('Month')
plt.ylabel('Sales')
plt.show()
```



Step 4 :- Create a bar chart

```
# calculate the total sales value grouped by product
labels = df.groupby('Product')['Sales'].sum()
```

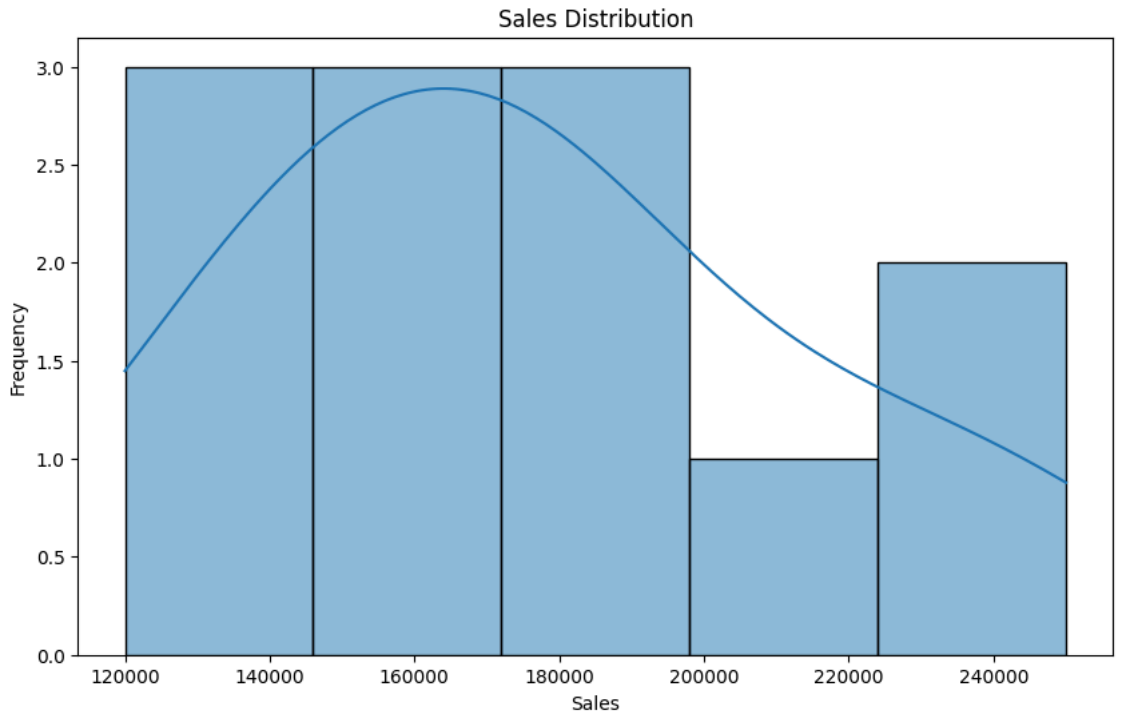
```
# create a bar chart using sales for product wise sales
plt.figure(figsize=(10,6))
bars = plt.bar(df['Product'],df['Sales'])
plt.title('Product wise sales')
plt.xlabel('Product')
plt.ylabel('Sales')
plt.bar_label(container=bars, labels=labels)
plt.show()
```



Step 5 :- Create a histogram

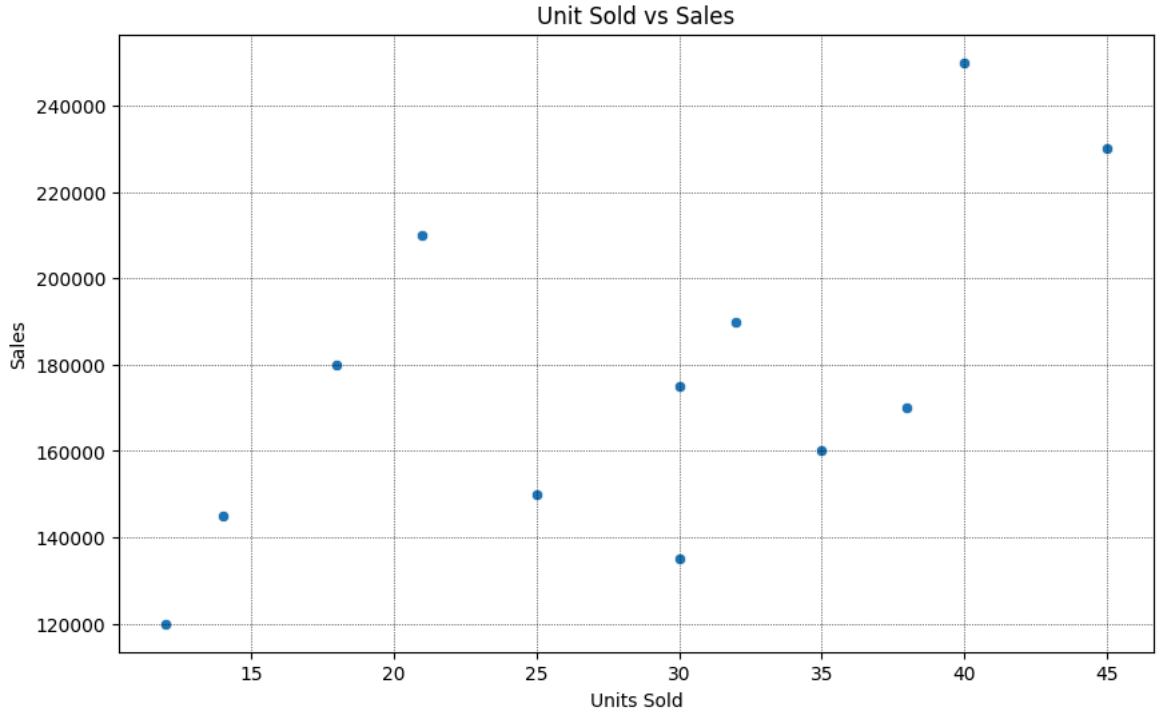
```
# create a histogram of the distribution of sales with kernel density estimate
plt.figure(figsize=(10,6))
sns.histplot(x = df['Sales'],kde=True)
```

```
plt.title('Sales Distribution')
plt.xlabel('Sales')
plt.ylabel('Frequency')
plt.show()
```



Step 6 :- Create a scatter plot

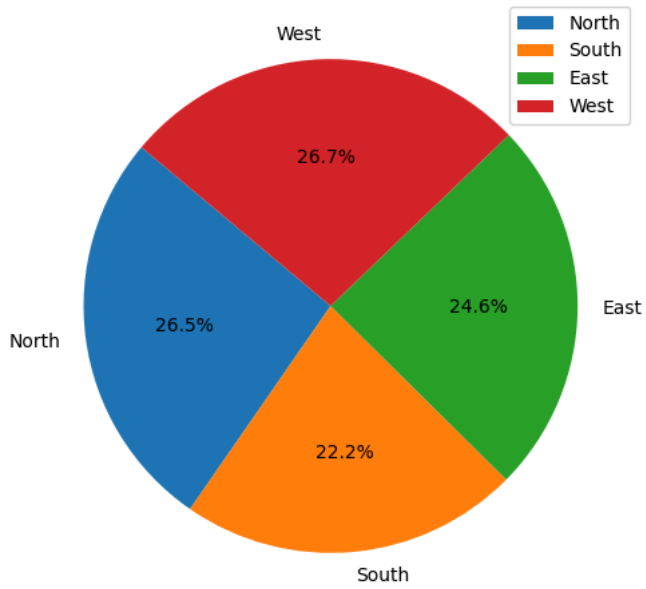
```
# create a scatter plot chart for units sold versus total sales .
plt.figure(figsize=(10,6))
sns.scatterplot(x = df['Units_Sold'], y = df['Sales'])
plt.grid(True, which='both', linestyle=':', linewidth=0.5, color='black')
plt.title("Unit Sold vs Sales")
plt.xlabel("Units Sold")
plt.ylabel("Sales")
plt.show()
```



Step 7 :- Create a pie chart

```
# calculate the total sales grouped by region
data = df.groupby('Region')['Sales'].sum()
```

```
# creating a pie chart regionwise sales distribution
plt.figure(figsize=(10,6))
plt.pie(data, labels= df['Region'].unique(), autopct='%1.1f%%', startangle=140)
plt.legend()
plt.show()
```



Step 8 :- Save the common dataset and download CSV

```
# export the dataframe to csv and download it to local storage
df.to_csv('sales_analysis_dataset.csv', index=False)
from google.colab import files
files.download('sales_analysis_dataset.csv')
```